Finding Probability Distribution of Function of Random Variables

Introduction

To determine the probability distribution for a function of n random variables, $Y_1, Y_2, \ldots, Y_n$, we must find the joint probability distribution for the random variables themselves.

Cont.

We generally assume that observations are obtained through random sampling. Random sampling from a finite population (sampling without replacement) results in dependent trials but that these **trials become essentially independent if the population is large when compared to the size of the sample.**

We will assume throughout the remainder of this chapter that populations are large in comparison to the sample size and consequently that the **random variables obtained through a random sample are in fact independent of one another.**

Cont.

Therefore, in the discrete case, the joint probability function for $Y_1, Y_2, \ldots, Y_n$, all sampled from the same population, is given by

$$p(y_1, y_2, \ldots, y_n) = p(y_1)p(y_2)\cdots p(y_n).$$

In the continuous case, the joint density function is

$$f(y_1, y_2, \ldots, y_n) = f(y_1)\, f(y_2)\cdots f(y_n).$$

Finding the Probability Distribution of a Function of Random Variables

Methods

- The Method of Distribution Functions
- The Method of Transformations
- The Method of Moments
- The Method of Multivariable Transformation (Jacobian)
- The Method of Order Statistics

The Method of Distribution Functions

Method

Let U be a function of the random variables $Y_1, Y_2, \dots, Y_n$.

1. Find the region $U = u$ in the $(y_1, y_2, \dots, y_n)$ space.
2. Find the region $U \leq u$.
3. Find $F_U(u) = P(U \leq u)$ by integrating $f(y_1, y_2, \dots, y_n)$ over the region $U \leq u$.
4. Find the density function $f_U(u)$ by differentiating $F_U(u)$. Thus,

$$f_U(u) = d\,FU(u)/du.$$

EXAMPLE 01

A process for refining sugar yields up to 1 ton of pure sugar per day, but the actual amount produced, Y , is a random variable because of machine breakdowns and other slowdowns. Suppose that Y has density function given by

$$f(y) = \begin{cases} 2y, & 0 \le y \le 1, \\ 0, & \text{elsewhere.} \end{cases}$$

The company is paid at the rate of \$300 per ton for the refined sugar, but it also has a fixed overhead cost of \$100 per day. Thus the daily profit, in hundreds of dollars, is U = 3Y − 1. Find the probability density function for U.

EXAMPLE 02

Let Y have probability density function given by

$$f_Y(y) = \begin{cases} \dfrac{y+1}{2}, & -1 \le y \le 1, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the density function for $U = Y^2$.

EXAMPLE 03

The amount of flour used per day by a bakery is a random variable Y that has an exponential distribution with mean equal to 4 tons. The cost of the flour is proportional to $U = 3Y + 1$.

a Find the probability density function for U.
b Use the answer in part (a) to find $E(U)$.

EXAMPLE 04

Suppose that two electronic components in the guidance system for a missile operate independently and that each has a length of life governed by the exponential distribution with mean 1 (with measurements in hundreds of hours). Find the

a probability density function for the average length of life of the two components.

b mean and variance of this average, using the answer in part (a). Check your answer by computing the mean and variance, using Theorems(Chapter1) .

The Method of Transformations

Method

1. Let $U = h(Y)$, where $h(y)$ is either an increasing or decreasing function of y for all y such that $f_Y(y) > 0$.
2. Find the inverse function, $y = h^{-1}(u)$.
3. Evaluate

$$\frac{dh^{-1}}{du} = \frac{d[h^{-1}(u)]}{du}$$

4. Find $f_U(u)$ by

$$f_{U(u)} = fy[h^{-1}(u)] \left|\frac{dh^{-1}}{du}\right|$$

EXAMPLE 05

In Example 1, we worked with a random variable Y (amount of sugar produced) with a density function given by

$$f_Y(y) = \begin{cases} 2y, & 0 \leq y \leq 1, \\ 0, & \text{elsewhere.} \end{cases}$$

We were interested in a new random variable (profit) given by U = 3Y − 1. Find the probability density function for U by the transformation method.

EXAMPLE 06

Let Y have the probability density function given by

$$f_Y(y) = \begin{cases} 2y, & 0 \le y \le 1, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the density function of U = −4Y + 3.

EXAMPLE 07

Let Y_1 and Y_2 have a joint density function given by

$$f(y_1, y_2) = \begin{cases} e^{-(y_1+y_2)}, & 0 \le y_1,\ 0 \le y_2, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the density function for $U = Y_1 + Y_2$.

EXAMPLE 08

A fluctuating electric current I may be considered a uniformly distributed random variable over the interval (9, 11). If this current flows through a 2 -ohm resistor, find the probability density function of the power $P = 2I^2$.